

12 (**long**) Consider the circuit in Fig. 1 comprising a voltage source V , a switch, a resistor R , an inductor L and a capacitor C . The switch is open at times $t < 0$ and closed for $t \geq 0$. The function $i(t)$ represents the current through the circuit and the function $v(t)$ represents the potential difference across the capacitor.

- (a) Explain why $i(t)$ and $v(t)$ satisfy the differential equations

$$i = C \frac{dv}{dt}$$

$$Ri + L \frac{di}{dt} + v = VH(t)$$

where $H(t)$ is the Heaviside step function, i.e. $H(t) = 0$ for $t < 0$ and $H(t) = 1$ for $t \geq 0$. [2]

- (b) Show that the Laplace transform $\bar{i}(s)$ of $i(t)$ satisfies

$$\bar{i}(s) = \frac{V/L}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

noting that $i(0) = 0$ A and $v(0) = 0$ V. [6]

- (c) Derive an expression for $i(t)$ when $LC = 1$ and $RC = 2.5$. [7]

Remark: the unit for LC is seconds squared and the unit for RC is seconds, but we omit units in order not to risk confusing units with the variable s used in the Laplace transform.

- (d) Derive an expression for $v(t)$ when $LC = 0.8$ and $RC = 0.8$. [8]

- (e) Given $LC = 1$, for what value of RC does $i(t) = \beta t e^{-at}$ for some constants a and β ? What are a and β ? [7]

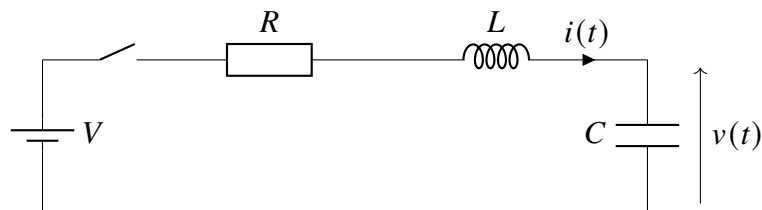


Fig. 1

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